SIMERJIT KAUR NATH

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MSc AI candidate at NTU (Jan 2027) who builds AI systems that hold up on real data. LLM applications and multi-agent pipelines, multimodal deep learning for city-scale digital twins at A*STAR, and efficient fine-tuning; first-author paper accepted at the SENTIRE Workshop, IEEE ICDM 2026. 2 years at NielsenIQ delivering research for global consumer brands and leading an 11-person team.

PUBLICATIONS

Nath, S. K., Cambria, E., Pen, H. *SenticLens: A Production Sentiment-Analysis Framework with Behavioral Auditing of LLM Affective Coherence.* SENTIRE Workshop, IEEE ICDM 2026 (IEEE ICDM proceedings; Scopus- and SCI-indexed). **Accepted (in press).**

EXPERIENCE

Graduate Researcher, SenticNet Lab, NTU *Aug 2025 – Present*

- Built **SenticLens**, an end-to-end sentiment and LLM evaluation framework ingesting live reviews from **three platforms** (Reddit, Apple App Store, Google Play), scoring polarity, intensity and emotion, with guardrails flagging inconsistent model outputs before results reach downstream analytics.
- Across **24,000 audited records**, showed models disagree with themselves **twice as often** on real reviews as on synthetic ones (38.16% vs 18.49%) and contradict themselves outright **about six times as often** (2.32% vs 0.37%), on every model and brand.
- Turned benchmark findings into **SenticCheck runtime rules** inside the live pipeline, not an offline report; modular, YAML-configured workflow (new brands with **zero code**) feeding engineering triage and business dashboards.

Research Intern, A*STAR Institute of High Performance Computing *Apr – Aug 2026*

Computational Sustainability

- Built and trained a **multimodal deep learning surrogate** fusing **four data modalities** (satellite imagery, street-level imagery, meteorological sensor streams and geospatial layers) to predict pedestrian-level mean radiant temperature at city scale for **digital-twin modelling**.
- Owned the **four-source** ingestion pipeline for urban climate observations and established the quality-control procedures the team now relies on.

Graduate Student Research Assistant, NTU School of Social Sciences (part-time) *Feb 2026 – Present*

- Built a local, offline LLM workflow for confidential interview transcripts with automated alignment and scoring, using LLM-assisted extraction, HDBSCAN with UMAP and Ward-linkage clustering.

Research Collaborator, LLM-Augmented Financial Signal Analysis *Jan 2026 – Present*

- Direct a **multi-agent LLM pipeline** for financial event interpretation, guiding experimental design, temporal-leakage control, catalyst validation and pipeline integrity, and co-supervising IIT Bombay undergraduate researchers.

Research Associate → Research Analyst → Team Lead, NielsenIQ, Mumbai, India *Jul 2023 – Jul 2025*

- Led an **11-member** ARC Europe task team to **97% feedback receipt**, a feedback score of 2.8 out of 3 and zero performance escalations.
- Delivered concept analysis and market research for **100+ FMCG launches** globally, including Reckitt and Nestlé, at 100% reported accuracy and a 3.9/4.0 project rating.

SELECTED TECHNICAL PROJECTS

- **RAG grounding audit** (open source, github.com/simer05/rag-grounding-audit, 2026): tested **3 open-weight LLMs** on **500 HotpotQA questions** under 6 controlled evidence conditions. Reordering the same passages flipped **8–14%** of answers between right and wrong while accuracy stayed flat, and models answered without evidence **26–40%** of the time.
- **AdaptLink, AI seller intelligence for Shopee** (Sea x OpenAI Codex Hackathon, Regional Finalist, team of 4, Jun 2026): came up with the product idea and built the seller-intelligence dashboard (TypeScript/Next.js), turning conversion, reviews, buyer questions, reply times and competitor prices into a product health score and up to **5 margin-safe actions**; designed a **6-agent** seller workflow with model routing.
- **CodeLah!, parameter-efficient code–NL translation** (Sep–Nov 2025): fine-tuned CodeGen-350M with LoRA for **100% syntax validity, 82.1% keyword coverage, 0.878 BERTScore** and a 40.8% gain over baseline.
- **Hardware-aware U-Net for Stable Diffusion** (Sep–Nov 2025): optimised an **860M-parameter** U-Net to **55% activation sparsity, 24% lower memory and 12.6% faster inference** at a 0.35-point FID change.

EDUCATION

Master of Science in Artificial Intelligence *Aug 2025 – Jan 2027 (expected)*

Nanyang Technological University, College of Computing and Data Science, Singapore

Bachelor of Science in Computer Science *Jul 2020 – Jun 2023*

SIES College, University of Mumbai, India · CGPA 9.63/10 (WES 3.86/4.00) · ***Silver Medalist***

AWARDS, LEADERSHIP AND SERVICE

Regional Finalist, Sea x OpenAI Codex Hackathon (2026) · Top 6 Finalist, American Express GenAI Hackathon (2025) · 2nd Place, TE Connectivity x AWS Hackathon (2025) · Bravo! Superstar, NielsenIQ/ARC Excellence Awards · Vice Chair, Operations and Logistics, TEDxNTU 2026 · Communication Coach, NTU CommCube (2026–Present): **95.6%** session satisfaction

TECHNICAL AND RESEARCH SKILLS

AI/ML: LLM evaluation and observability, guardrails, behavioural auditing, prompt engineering, supervised fine-tuning (SFT) with LoRA/QLoRA/PEFT, quantisation and 4-bit inference, deep learning, multimodal learning, NLP, sentiment and emotion analysis, diffusion models, physics-informed ML, time-series analysis, inference optimisation

ML systems: local and self-hosted deployment, multi-agent and agentic systems, LangGraph, sim-to-real, surrogate and reduced-order modelling for digital twins, experimental design, temporal-leakage control, responsible AI, reproducibility

AI coding tools: Claude Code, OpenAI Codex, Google Antigravity, Cursor, GitHub Copilot; hands-on with Claude, GPT, Gemini, DeepSeek, GLM (Z.ai), Grok and open-weight models

Tools: Python, PyTorch, TensorFlow, Hugging Face, scikit-learn, pandas, SQL, Git, Docker, AWS, Linux/HPC, Tableau, Power BI, Advanced Excel